Juniper

CLI coding assistant | Project report

Group Project 2 - 15 October 2026

1. System and design decisions

Juniper accepts natural-language coding tasks in a terminal. Its agent loop asks an LLM what to do, executes validated MCP tools, observes results, and streams its answer. It can edit files, run Python scripts/tests, and retrieve documentation. Groq, OpenAI, and Ollama share the provider interface. Historical Groq comparisons are retained; pages 11-12 report completed OpenAI follow-up evaluations.

Design decision	Reason and implemented behavior
Separate providers from the agent	A shared streaming interface lets the same agent and tools run with Groq, OpenAI, or Ollama. Model comparisons reuse the execution pipeline.
Discover tools through MCP	The required filesystem, hosted DeepWiki, and custom RAG servers are joined by a Python-check server. All tools are dynamically discovered through MCP.
Validate before execution	The dispatcher checks names and argument schemas. Confirmation is the default. Python checks validate the target path, use no shell, omit credential environment variables, and bound duration/output; executed code is trusted, not OS-sandboxed.
Keep retrieval local and traceable	Chroma stores document embeddings. Returned snippets retain source, page, and chunk references. Cross-encoder reranking improves ordering within retrieved candidates.
Bound cloud requests and recovery	Cloud adapters compact history and expose bounded retries after errors. Automatic Groq pacing has been removed. Checkpoints allow later resume; full local history is retained.

Evidence status

The original Groq batch remains quota-limited. A separate OpenAI follow-up completed all eight coding slots and all 20 RAG answers (pages 11-12). These experiments retain failures and do not fill Groq slots with a different provider. Passing local tests alone is not proof of autonomous coding.

Report guide: historical model comparison p. 2; historical RAG results p. 3; reflection p. 4; original planning p. 5; updated architecture p. 6; state/sequence diagrams pp. 7-10; completed OpenAI evaluations pp. 11-12.

2. Two LLMs on the same coding tasks

Models: **openai/gpt-oss-20b** and **openai/gpt-oss-120b**, both through Groq. The baseline used four identical task definitions, three repetitions per model, fresh conversations, and isolated copies of the same initial files: 24 scored attempts. Only workspace paths differed in task prompts. All runs used the production agent, all three MCP servers (18 discovered tools), auto execution, up to eight iterations, and a 180-second task deadline. Baseline generation settings were SDK defaults.

A code-check pass means saved code met the evaluator's behavioral checks. An end-to-end pass additionally requires successful read/edit tools, required retrieval tools for the documentation task, normal completion, and no terminal agent error. Checks run outside the model's accessible fixture root.

Identical task	20B end-to-end	120B end-to-end
Fix mean calculation; raise ValueError for empty input	1 / 3	1 / 3
Extract unique ASCII words in first-seen order	3 / 3	0 / 3
Refactor sales totals into a shared helper	0 / 3	0 / 3
Implement UTF-8 greeting writer using retrieved docs	0 / 3	0 / 3
Total end-to-end passes	4 / 12 (33.3%)	1 / 12 (8.3%)
Total code-check passes	4 / 12	3 / 12

Insights from the baseline

20B completed the ASCII-word feature on all three attempts; 120B repeatedly requested unqualified tool names. Conversely, 120B produced two correct sales refactors, but subsequent requests exceeded the account's 8,000-token-per-minute allowance, preventing end-to-end passes. Both models fixed the mean only on their third attempt. Tool use, recovery, and quota constraints strongly affected outcomes; model size alone did not predict success.

Subsequent diagnostics and current limits

Two intermediate eight-run pilots scored **3/8** and **2/8**. The second had zero dispatched tool errors, but still failed on limits, Windows indentation, and ASCII/Unicode logic. A later free-tier batch with pacing, checkpoints, and syntax checks passed the bug-fix task with **both models**. The next slot hit the daily cap before tool use, and the runner paused: **2/3 attempted, 2/8 planned**, with five slots unattempted. This remains a partial retest, not a clean overall improvement estimate or general model ranking.

Sources: evaluations/model_comparison/results/2026-10-15/, results/2026-10-15-fixes/, and results/2026-10-15-free-tier/. Each retains its own report, traces, source files, and metadata.

3. Advanced RAG and insights

The chosen technique is **cross-encoder reranking**. The corpus contains FastMCP documentation and Python, JavaScript, and Java PDFs. Text is chunked and embedded with **all-MiniLM-L6-v2**; Chroma indexes 10,473 chunks. At query time, cosine search finds candidates, then **cross-encoder/ms-marco-MiniLM-L6-v2** jointly scores each query/passage pair. The server returns original snippets and references in the new order.

Why this technique

A semantically related passage can rank highly without answering the question. Reranking adds a query-aware relevance step while reusing the existing vectors. The MCP tool defaults to reranking; cosine mode remains available as the baseline. Cosine similarity and cross-encoder logits are distinct scores, and neither is a confidence percentage.

Paired experiment: relevance, answers, and grounding

Ten fixed questions cover all four documents: eight intended answerable questions, a Path.write_text coverage probe, and an unrelated negative control. Both modes use the same production stdio MCP server and index, top_k=5, and no source filter. Reranking scores 20 cosine candidates. Generation uses the same Groq 20B adapter, evidence-only prompt, temperature 0, and SDK-default text-only reasoning; only the five supplied passages differ. Mode order alternates. This isolates context quality, not autonomous tool selection.

Metric	Cosine baseline	Cross-encoder
Direct evidence at rank 1 (8 queries)	25.0%	50.0%
Precision@5 (directly useful passages)	25.0%	35.0%
Mean reciprocal rank@5	0.400	0.688
Correct reference facts (5 answer pairs)	8/10	10/10
Fully grounded answers	2/5	4/5
Median warm retrieval time (10 queries)	0.021 s	0.430 s

What we learned

Reranking brought a complete FastMCP tool example into the final context; the baseline answer invented unsupported decorator arguments. Python break moved above a JavaScript passage. Gains were uneven: the reranked Java finally result was JavaScript, and the Python continue context lost a useful Python definition. Neither mode retrieved the required Path.write_text API detail. Source scope and corpus coverage remain important.

Groq's daily limit stopped generation after five complete pairs (10/20 planned answers); remaining answers, including controls, were not tested. All 100 final passages and 200 recorded candidates matched prepared text/metadata. A single assessor graded passages: 2 directly supports a reference fact in the requested language, 1 is related background, 0 is irrelevant/wrong-language. Precision/MRR count grade 2; answer correctness and full claim support are separate. Small samples, no blinded second rater, one generation per condition, and approximate search limit generalization. Warm timings exclude model loading and are not a controlled latency benchmark.

Reproduce: evaluations/rag_evaluation/README.md and results/2026-10-15/. Method reference: NirDiamant RAG_Techniques - Reranking (accessed 15 October 2026). Answer comparisons do not establish improved autonomous coding pass rates.

4. Reflection and supporting evidence

What worked

The provider-neutral loop connected the required tools and retained code/trace evidence. Several coding attempts saved correct implementations. The completed OpenAI follow-up passed seven of eight coding slots and generated all 20 RAG answers. Current regression and type-check results are recorded with the follow-up evidence.

What did not work

Incorrect tool names, null arguments, malformed output, long histories, and quotas interrupted tasks. Windows line-ending mismatches could damage indentation. Tool success did not imply correct code: a later 120B feature mishandled Unicode when enforcing ASCII words. Retrieval sometimes returned related but insufficient evidence; scanned PDF pages limited corpus coverage.

Changes made and what we would improve

Implemented fixes include tool aliases, null cleanup, bounded recovery, context compaction, Windows-edit normalization, and the correct RAG interpreter. Checkpoints preserve runs for resume; current Groq requests are no longer automatically paced. Python edits receive policy-controlled reads and syntax checks; failures block normal completion. Syntax checks do not prove behavior.

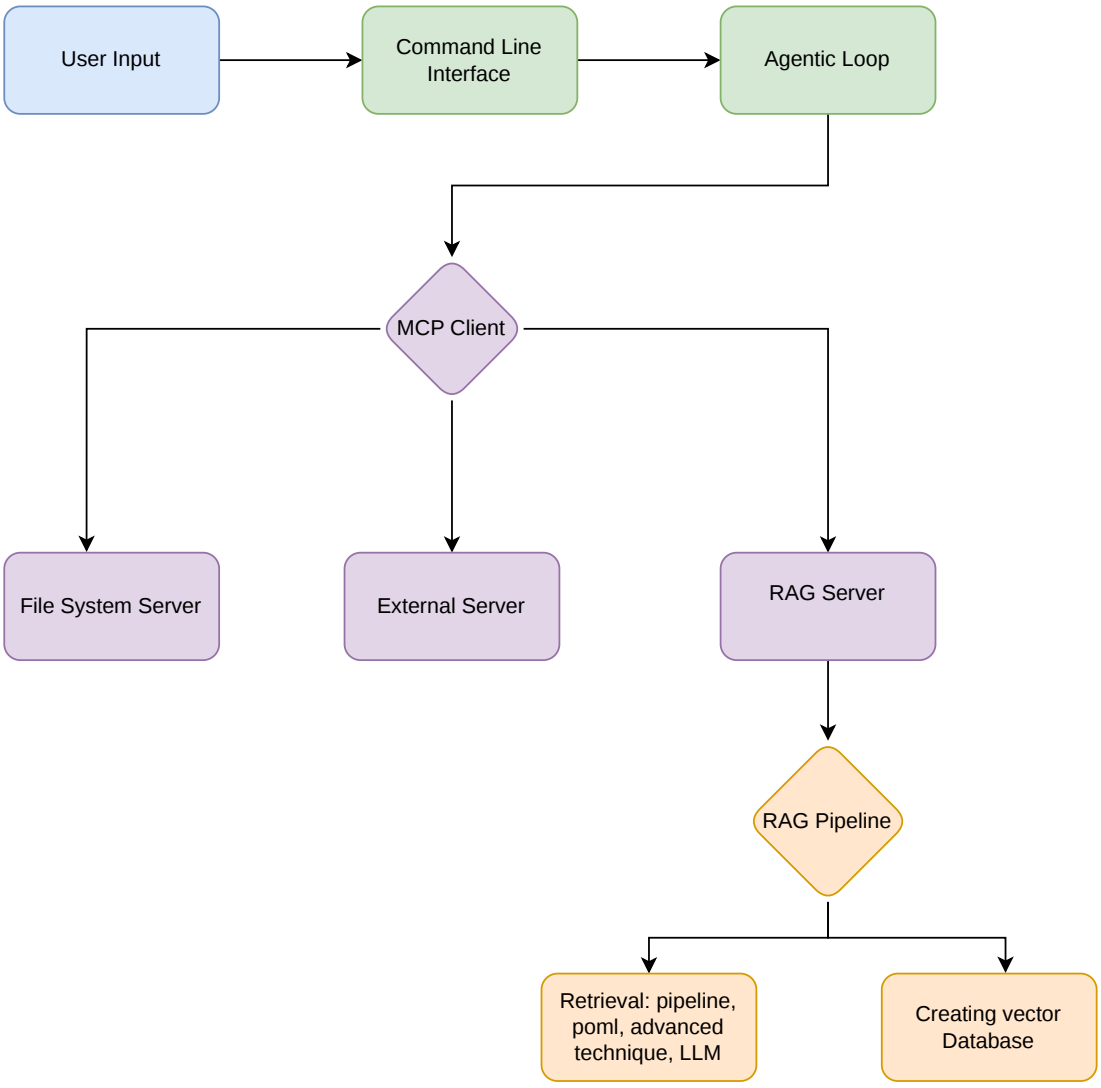
A separate MCP tool runs trusted Python scripts or pytest with approval, exit status, timeouts, and bounded output. It is a building block for behavioral checks, not proof of adequate coverage. Pages 11-12 complete new OpenAI coding and paired-answer experiments. Next we would improve corpus coverage/source filtering and use repeated generations and independent blinded raters. Provider quotas remain outside the project.

Evidence index (paths relative to repository root)

Evidence	Location
24-run baseline and saved code	evaluations/model_comparison/results/2026-10-15/
Fixes, 16 diagnostic runs, final-policy hashes	evaluations/model_comparison/results/2026-10-15-fixes/
Measured RAG comparison and answer judgments	evaluations/rag_evaluation/README.md; results/2026-10-15/
Free-tier controls and syntax validation	evaluations/model_comparison/FREE_TIER_TESTING_NOTES.md
Original planning architecture (p. 5)	init-architecture.drawio2.pdf
Current structure and configuration (p. 6)	docs/architecture.md; config/mcp.json; src/coding_assistant/; rag_server/

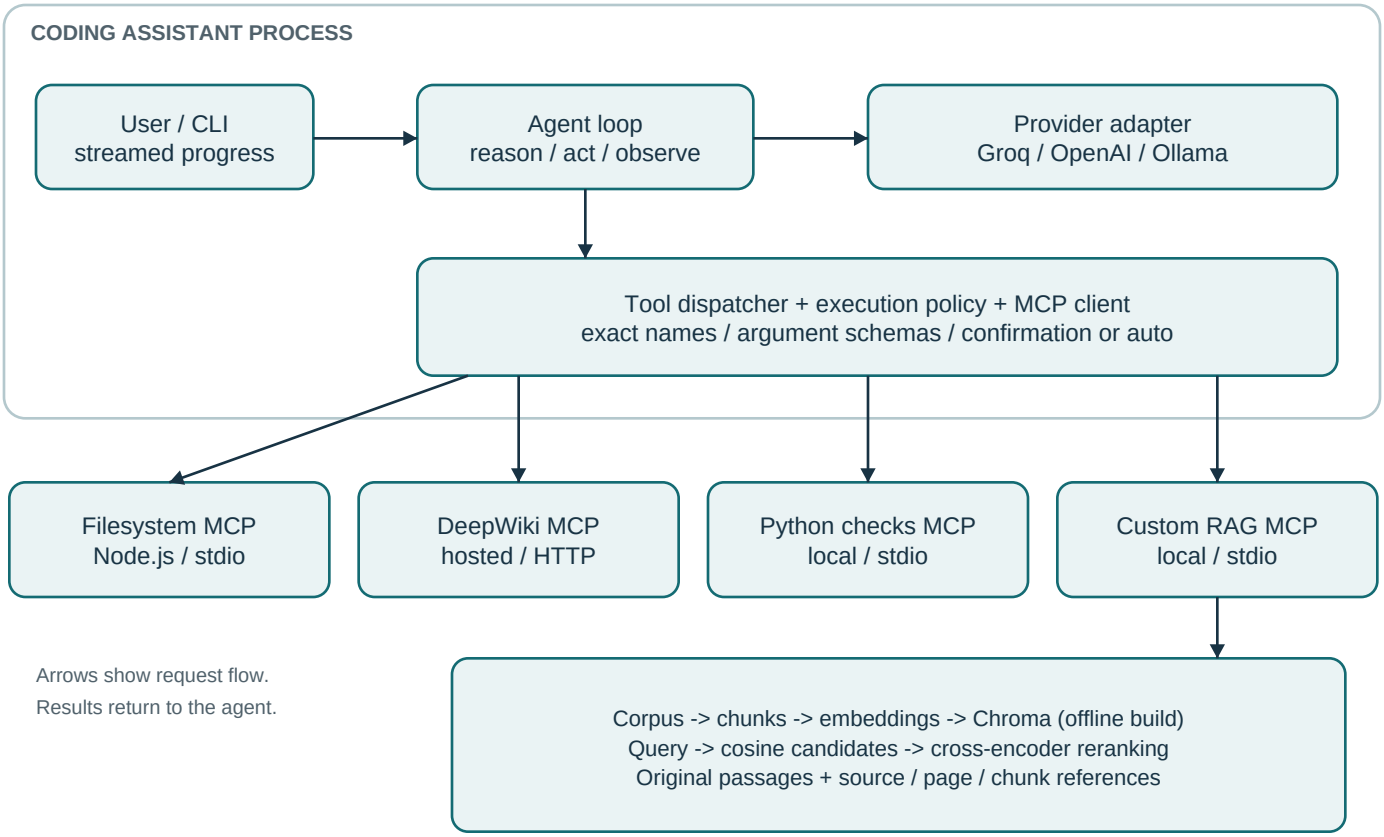
The coding baseline evaluated commit 6e4735d; later fixes have retained source hashes. The RAG experiment records base commit 031edeb plus executed source hashes. Later command-tool additions were not part of those coding trials. The video demo remains a separate deliverable for the team. Page 5 reproduces the original planning PDF unchanged.

Initial Architecture



5. Updated architecture

The original planning diagram appears on page 5. This diagram shows the final system, including reliability changes and the Python-check server. It preserves the planned CLI, agent, MCP client, and three required server roles.



What changed from the plan, and why

Model boundary made explicit. Groq, OpenAI, and Ollama adapters sit outside the agent's coordination logic so models can change without replacing the tool pipeline. The planning diagram did not show this boundary.

Execution checks added explicitly. Discovered tool names, schemas, and confirmation policy sit before MCP calls. Provider-side aliases, context budgeting, and retries address the failures observed during evaluation.

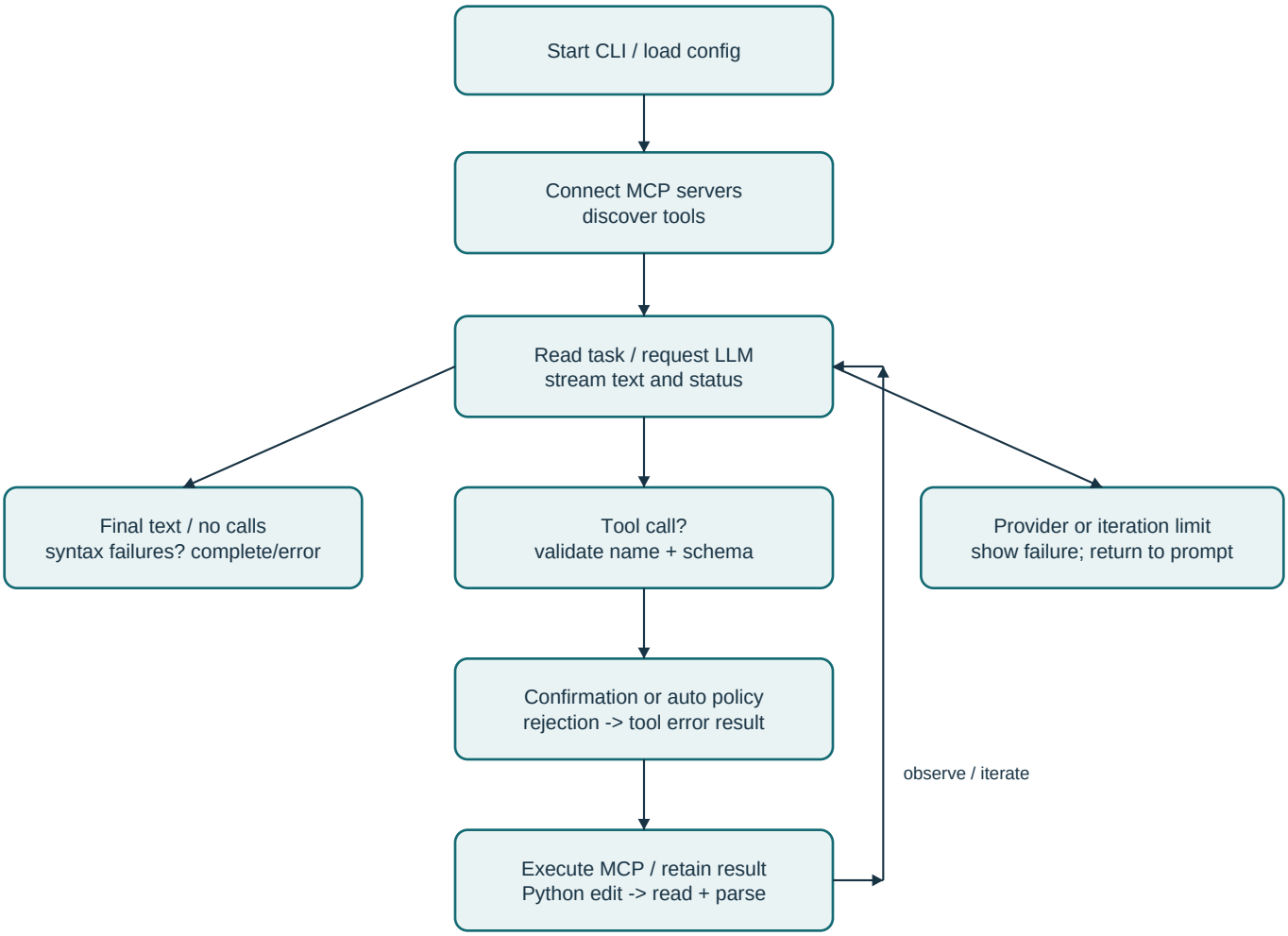
RAG design made concrete. The plan's generic retrieval/vector-database branches became an offline ingestion/index build and a query-time cosine-plus-cross-encoder pipeline. Answer generation remains in the main agent/provider; the RAG server returns evidence. Chroma and reranking were chosen to reuse vectors and improve passage ordering.

Deployment boundaries clarified. Filesystem and RAG are local stdio processes; DeepWiki is hosted over HTTP. The RAG executable follows the running Python interpreter to avoid missing dependencies. These refinements explain how the original architecture became executable.

Command execution added. A separate local Python-check MCP server runs trusted scripts/tests with per-call policy, target validation, time limits, credential-environment omission, and bounded returned output. This fills the assignment's command-execution capability without adding a shell interface. It is not an OS sandbox.

6. Task state diagram

The state flow combines user input, model decisions, validation, approval, MCP execution, feedback, and completion. See docs/diagrams.md for editable Mermaid sources and error branches.



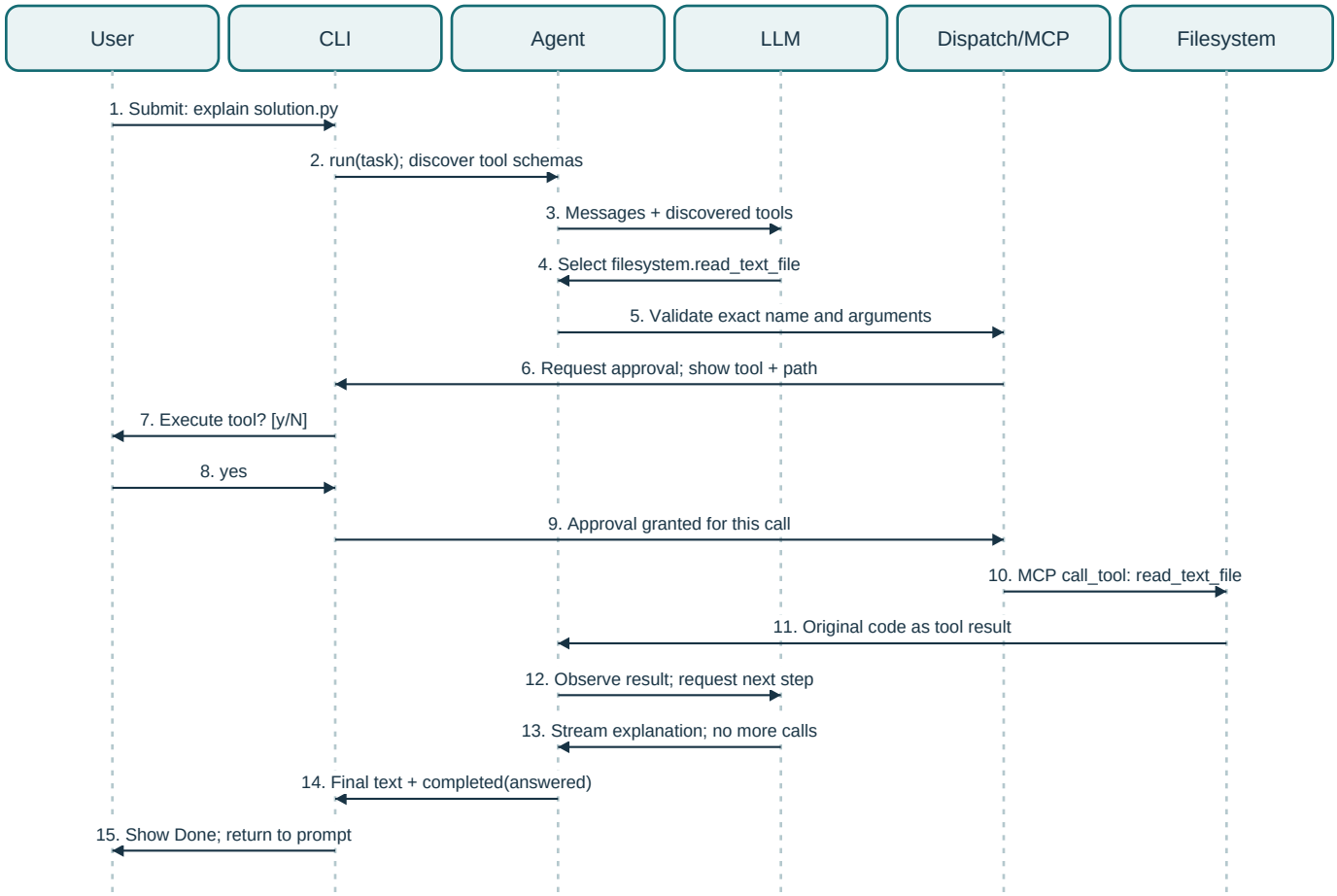
Completed or failed task -> next prompt. /exit or interruption -> close clients and exit.

Invalid calls return errors without execution. Groq retries are bounded inside its adapter.

The CLI can run many tasks in one session. One-shot mode exits after its task. All clients close on exit. The LLM sees full tool results through the agent; the Groq adapter may compact the request view. Python syntax feedback is added after writes, while optional behavioral execution is a distinct checks.run_python_check tool call.

Inspect existing code

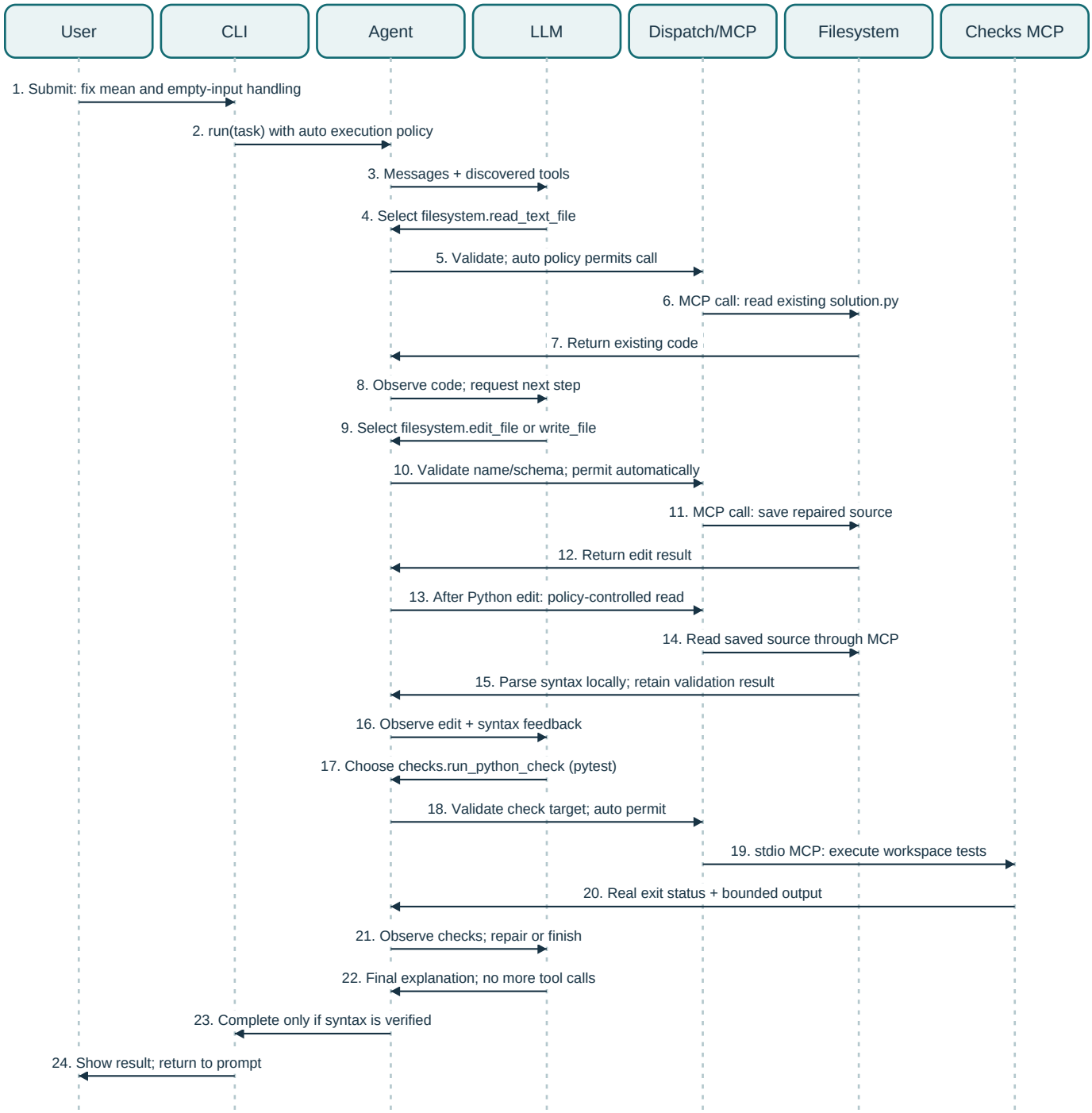
Confirmation mode | operation 1



If approval is rejected, no filesystem call occurs. The rejection is returned to the LLM as a tool error, so it can explain or revise its plan.

Repair a Python bug

Automatic mode | operation 2

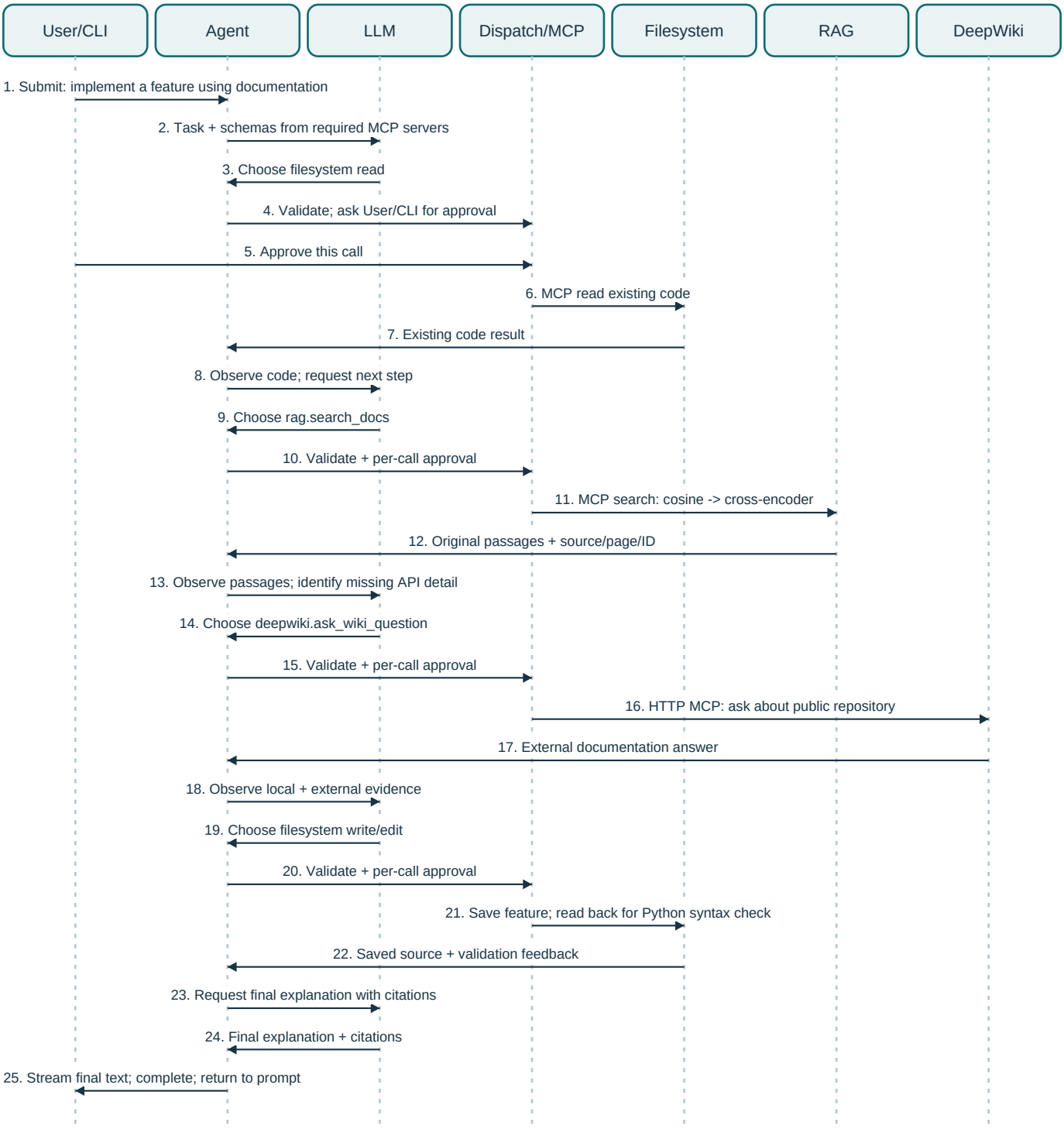


Syntax errors feed back for repair and block normal completion until resolved. The model can request the separate Python-check tool to run behavioral tests; those tests establish only what they check. Historical coding evaluation checks ran separately outside the model's writable fixture.

The optional Python-check call shown here was added after the recorded coding comparison. Test completeness and assertions determine what a passing check means.

Implement a documented feature

Confirmation mode | operation 3



Each valid call, including syntax-check reads, requests approval in confirmation mode. The model chooses tools and their order; this is an implementation walkthrough, not a claim that every live task succeeds. If local evidence is insufficient, retrieve another source or acknowledge the gap.

7. Completed OpenAI coding evaluation

Follow-up to the quota-limited Groq trials | 15 October 2026

All eight planned task/model slots completed with real OpenAI API calls. Models were fixed snapshots **gpt-4.1-mini-2025-04-14** and **gpt-4.1-2025-04-14**. Both used the same four task definitions, fresh workspaces, production agent/tool pipeline, auto execution, eight model turns, and a 180-second deadline per task. All four configured MCP servers exposed 19 tools.

Same task	GPT-4.1 mini	GPT-4.1
Repair arithmetic mean	Pass	Pass
Unique lowercase ASCII words	Fail	Pass
Share region aggregation	Pass	Pass
UTF-8 greeting using retrieved docs	Pass	Pass
Total machine-checked passes	3/4	4/4

What failed, and what the pass counts mean

Mini lowercased characters before checking ASCII membership, so the Kelvin sign became an accepted letter. The hidden Unicode case failed despite valid syntax and normal completion. GPT-4.1 passed the same checks. Mini's documentation implementation passed, but its explanation falsely claimed local RAG supported Path.write_text; its successful external call only listed wiki pages after two failed content requests. GPT-4.1 retrieved the API details through DeepWiki and acknowledged the local coverage gap. Machine coding passes therefore do not establish fully grounded explanations.

Method and retained limits

The existing OpenAI adapter uses streamed Chat Completions, SDK-default temperature, 4,096 completion tokens, a 30,000-byte context view, and at most two bounded retries. One trial per task/model is descriptive, not a statistical ranking. Provider, settings, date, source hashes, code, and full traces are retained. Differences from the historical Groq configuration prevent attributing the improvement solely to model choice.

The first OpenAI batch stopped on a Windows grader decoding error after two scored passes. Its scored runs and interrupted workspace remain in results/2026-10-15-openai/. The grader now explicitly uses UTF-8 in both isolated child and parent; the complete replacement batch above starts fresh. No failed model task was retried to improve the score. No Groq API calls were made.

Evidence: evaluations/model_comparison/results/2026-10-15-openai-complete/ (metadata, results, code, source_snapshot, TEST_RESULTS.md, and OPENAI_FOLLOWUP.md).

8. Completed OpenAI RAG answers

Ten questions, two contexts each, 20 fresh API answers

All 20 planned answers were generated with **gpt-4.1-mini-2025-04-14**. Both modes used the original evidence-only prompt and exact retained top-five passages, with alternating mode order. Retrieval.json and queries.json are byte-identical to the historical experiment and verified by hash. All answers were generated anew; no OpenAI answer was paired with a Groq answer. Retrieval quality and timings are historical, not remeasured.

OpenAI answer measure	Cosine	Rerank
Completed answers, including controls	10/10	10/10
Correct facts, 8 answerable questions	12/16	13/16
Fully grounded answers, 8 questions	4/8	6/8
Answers with valid citation IDs	6/8	7/8
Appropriate abstention, 2 coverage controls	2/2	2/2

What the complete answer set reveals

Reranking supplied a usable FastMCP registration example and CLI startup command. Its tool-registration answer nevertheless abbreviated citation IDs, so correct code did not count as fully grounded. Cosine included a malformed stdio citation and an invented [55] reference in the finally answer. Grouped full-length IDs are accepted by the updated scorer; missing, shortened, and out-of-context IDs are not silently credited.

Both modes declined the unsupported Apollo-menu and Path.write_text questions. Reranking still over-abstained on Java inheritance despite an implements-clause passage. Cosine supplied the correct Java keywords using knowledge absent from its context. Retrieval relevance, reference-fact correctness, and evidence grounding remain separate measures.

Review and limits

Codex reviewed all 20 answers against their supplied passages; this is one assessor, without blinding or independent validation. Frozen retrieval grades were reused only after matching evidence hashes; answer judgments are new. The follow-up explicitly requires the after-loop fact to be stated, while the older assessor accepted implication. Do not use the old/new totals as a model ranking. One generation per context, SDK-default sampling, small authored questions, and inherited corpus gaps limit generalization.

Evidence: evaluations/rag_evaluation/results/2026-10-15-openai/ (answers, judgments, metrics, metadata, state, and source_snapshot). Reproduce with python -m evaluations.rag_evaluation.openai_answers --output NEW_DIR; recompute with python -m evaluations.rag_evaluation.score RESULT_DIR. The original quota-limited Groq evidence is retained unchanged.